

Multi-Task Tennis Stroke Biomechanics Analysis Using MediaPipe Pose

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Abstract

We present a multi-task pipeline for tennis stroke biomechanics analysis from monocular RGB video, extending the pose-based stroke recognition of Shimizu et al. [6] with two novel learned tasks—stroke direction prediction and posture quality assessment—plus a rule-based shot selection feedback layer. Stroke boundaries are detected automatically via a weighted joint velocity score $st = 0.5 \cdot v_{\text{wrist}} + 0.3 \cdot m_{\text{elbow}} + 0.2 \cdot m_{\text{shoulder}}$, eliminating manual pre-annotation. Pose features are extracted using MediaPipe Pose Landmarker (33 landmarks, metric world coordinates), and each stroke is represented as a sequence $\mathbf{X} \in \mathbb{R}^{30 \times 39}$ fed to *TennisTransformerGPU*—a 564,103-parameter transformer encoder (4 layers, 4 heads, $d = 128$) with three parallel task heads. Trained on 1,281 annotated strokes from 7 professional players and 1 amateur across 11 videos, the model achieves 83.7% stroke-type accuracy, 61.9% direction accuracy, and 62.6% posture accuracy under random 80/20 evaluation. A cross-player experiment (professional $\rightarrow$ amateur) yields 82.9% stroke-type accuracy with a 0.8% drop; direction collapses to the majority class under cross-player evaluation, indicating that genuine direction transfer is not achieved. An ablation study confirms that world coordinates are critical to this generalization: replacing them with image landmarks drops cross-player stroke-type accuracy from 83% to 47% and direction accuracy from 68% to 21%. The full pipeline runs on Kaggle free-tier hardware (T4 GPU) and is openly reproducible.

Keywords

tennis stroke analysis, pose estimation, multi-task learning, transformer, biomechanics, sports analytics, MediaPipe, cross-player generalization

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1 Introduction

Automated analysis of athletic technique has long been a goal of sports computer vision. In tennis, stroke mechanics directly govern match outcomes: subtle differences in racket path, body rotation, and weight transfer separate elite from

recreational play. Yet most computational approaches focus on ball tracking [3], court detection, or coarse action recognition, leaving biomechanical correctness largely unaddressed.

Shimizu et al. [6] introduced a pose-driven pipeline for tennis stroke recognition using OpenPose and a CNN/SVM classifier, achieving ~ 85 – 90% stroke-type accuracy on broadcast footage. That system identifies *what* stroke was played, not *how well*, in *which direction*, or *whether* it was the right tactical choice. We address this gap with a fully automated, multi-task biomechanics pipeline. Our contributions are:

- (1) **Automated stroke segmentation.** A velocity-based detector using weighted wrist, elbow, and shoulder motion, eliminating manual pre-annotation.
- (2) **Richer pose representation.** MediaPipe Pose Landmarker provides 33 landmarks in metric world coordinates (hip-origin, camera-independent), adding depth and visibility scores unavailable in the prior framework. An ablation study (Section 4.1) shows world coordinates yield up to $+47\%$ accuracy gain over image landmarks under cross-player evaluation.
- (3) **Two novel learned tasks plus a feedback layer.** Dedicated heads for shot direction prediction and posture quality assessment, plus a rule-based coaching feedback layer requiring no additional labels.
- (4) **Cross-player generalization.** Seven professionals and one amateur in the dataset, enabling a pro-to-amateur transfer experiment directly relevant to coaching applications.
- (5) **Accessibility.** Full pipeline runs on Kaggle free-tier T4 GPU hardware, openly reproducible.

2 Related Work

2.1 Pose Estimation in Sports

Pose estimation has been widely applied in sports, from gymnastics action quality [5] to gait analysis. OpenPose [1] enabled per-frame 2D landmark extraction; MediaPipe Pose [4] extends this with metric 3D world coordinates without depth sensors. Faulkner and Dick [2] applied pose to tennis event detection but did not address biomechanical coaching tasks.

2.2 Stroke Recognition and Classification

Shimizu et al. [6] used CNN-extracted pose features with an SVM classifier on large broadcast datasets. Mora and Knottenbelt [7] employed optical flow and SVMs; more recent work uses 3D CNNs and RNNs. Our transformer encoder attends

jointly across the 30-frame stroke window and simultaneously optimizes three task heads.

2.3 Multi-Task Learning in Sports Analytics

MTL has shown consistent benefits for injury risk, basketball shot quality, and swimming stroke symmetry [8]. No prior work, to our knowledge, applies MTL to tennis biomechanics with the combination of tasks we introduce.

3 Dataset Creation

3.1 Video Collection

We collected 11 videos across two player categories:

- **Professional (7):** Alcaraz, Federer, Djokovic, Nadal, Zverev, Sinner, Wawrinka — sourced from YouTube practice footage.
- **Amateur (1):** One recreational player, self-recorded, to enable a cross-player generalization study.

Videos range from 30 to 60 FPS.

3.2 Annotations

Strokes were manually annotated along three axes: (1) **stroke type** (forehand/backhand); (2) **shot direction** (cross-court/down-the-line/center); (3) **posture quality** (good/bad), assessed on hip-shoulder separation, balanced stance, and racket preparation timing. All annotations were produced by a single annotator (the author); inter-rater reliability was not measured, which is a limitation of the posture task in particular. Table 1 reports the full class distribution across 1,281 instances. Direction labels are notably imbalanced (62% center), consistent with the dominance of neutral-stance rally footage. Stroke type is 77% forehand and posture is 63%/37% good/bad, motivating the class-weighted loss described in Section 4.4.2.

Table 1: Class distribution across all annotation axes (1,281 total).

Task	Class	Count
Direction	Center	785
	Cross-court	297
	Down-the-line	199
Stroke Type	Forehand	982
	Backhand	299
Posture Quality	Good	812
	Bad	469

3.3 Frame Processing

All frames are processed without subsampling. With GPU acceleration (Kaggle T4), MediaPipe pose extraction runs on every frame regardless of source frame rate, so no temporal information is discarded.

4 Methodology

Our pipeline consists of four sequential stages, illustrated in Figure 1.



Figure 1: End-to-end pipeline overview. Input videos (4 amateur-recorded + 7 professional YouTube + 2 multi-player) pass through MediaPipe pose extraction and automated stroke detection, manual annotation, and label encoding. A sequence builder constructs fixed-length stroke windows fed to *TennisTransformerGPU* (d=128, 4 layers, 4 heads). Results reported under random 80/20 and cross-player protocols.

4.1 Stage 1: Pose Extraction

MediaPipe Pose Landmarker (full model, float16) extracts per-frame pose:

$$\mathbf{f}_k = x_k, y_k, z_k, v_k, wx_k, wy_k, wz_k$$

where x_k, y_k, z_k are image-normalized coordinates, v_k is visibility, and wx_k, wy_k, wz_k are metric world coordinates in a hip-origin, camera-independent frame. We select 13 landmarks (wrists, elbows, shoulders, hips, knees, ankles), yielding a per-frame feature vector of $13 \times 3 = 39$ dimensions (world coordinates only). Figure 2 shows a representative result.

To validate the choice of world coordinates over image landmarks, we conducted a controlled ablation study training identical model configurations on each coordinate type separately. Table 2 reports the results.

Table 2: Ablation: image landmarks vs. world landmarks across all tasks and evaluation protocols. World coordinates consistently outperform image landmarks, with the largest gain on direction prediction (cross-player: +47%).

Task (Protocol)	Image	World	Δ
Direction (random)	43%	62%	+19%
Stroke type (random)	80%	84%	+4%
Posture (random)	53%	63%	+10%
Direction (cross-player)	21%	68%	+47%
Stroke type (cross-player)	47%	83%	+36%
Posture (cross-player)	39%	63%	+24%

World coordinates improve performance across all tasks, with the largest gains under cross-player evaluation: direction rises from 21% to 68% (+47%), stroke type from 47% to 83% (+36%), and posture from 39% to 63% (+24%). These results confirm the core hypothesis—image landmarks encode camera viewpoint and player position, making them brittle when the test player differs from training. World coordinates, being metric and hip-origin, factor out these confounds. Notably, the 68% cross-player direction figure reflects majority-class collapse rather than genuine directional transfer (see Section 5.2); the +47% still captures the benefit of world coordinates in avoiding the near-random 21% obtained with image landmarks. The gap is smaller under random split (where train and test share the same players), confirming that the benefit is specifically in generalization rather than in-distribution fitting.

Stroke 889 | nadal | Rafael Nadal vs Tommy Paul INTENSE Practice Washington Citi Open Court Level Vi
Frame: 26676 | Time: 444.60s | Pose detected



Figure 2: MediaPipe Pose Landmarker on a Nadal forehand (Washington Citi Open). 33 landmarks extracted per frame with image and world coordinates. Pose is reliably detected during dynamic mid-stroke configurations under broadcast conditions.

4.2 Stage 2: Automated Stroke Segmentation

We introduce a velocity-based segmentation algorithm replacing the manual pre-annotation of prior work [6]. For each

frame t :

$$st = 0.5 \cdot v_{\text{wrist}} t + 0.3 \cdot m_{\text{elbow}} t + 0.2 \cdot m_{\text{shoulder}} t$$

where v_{wrist} is dominant wrist velocity in world space, and m_{elbow} , m_{shoulder} are frame-to-frame displacement magnitudes. Local peaks exceeding a player-adaptive threshold are stroke events:

$$\tau = \begin{cases} 0.07\text{--}0.08 & \text{professional players} \\ 0.04 & \text{amateur player} \end{cases}$$

A minimum 40-frame inter-stroke gap suppresses double detections. Each peak defines a ± 15 -frame window (30 frames total).

4.3 Stage 3: Sequence Feature Construction

Each stroke is a fixed-length sequence $\mathbf{X} \in \mathbb{R}^{30 \times 39}$. Sequences shorter than 30 frames are zero-padded; longer ones are center-cropped.

4.4 Stage 4: Multi-Task Transformer

4.4.1 Architecture. *TennisTransformerGPU* is a transformer encoder with three parallel task heads, summarised in Table 3. Hyperparameters ($d_{\text{model}}=128$, $\text{num_layers}=4$, $\text{dropout}=0.3$) reflect iterative tuning from a smaller prototype ($d_{\text{model}}=64$, 2 layers) as dataset size grew with GPU access.

Table 3: *TennisTransformerGPU* architecture (564,103 parameters).

Component	Configuration
Input projection	Linear 39 $\rightarrow$ 128
Positional encoding	Learned, shape 1, 30, 128
Transformer encoder	4 layers, 4 heads, FFN=256, drop=0.3
Pooling	AdaptiveAvgPool1d
Direction head	128 $\rightarrow$ 64 $\rightarrow$ ReLU $\rightarrow$ 64 $\rightarrow$ 3
Stroke type head	128 $\rightarrow$ 64 $\rightarrow$ ReLU $\rightarrow$ 64 $\rightarrow$ 2
Posture head	128 $\rightarrow$ 64 $\rightarrow$ ReLU $\rightarrow$ 64 $\rightarrow$ 2
Total parameters	564,103

The three learned heads cover stroke type ($n_c = 2$), direction ($n_c = 3$), and posture ($n_c = 2$). Shot selection feedback is a deterministic post-processing layer mapping direction and posture outputs to one of four coaching messages via a hand-crafted decision table, requiring no additional labels.

The complete PyTorch implementation is given in Listing 1.

```

1 class TennisTransformerGPU(nn.Module):
2     def __init__(self, input_dim=39, d_model=128,
3                   nhead=4, num_layers=4, dropout=0.3,
4                   n_directions=3, n_stroke_types=2,
5                   n_posture=2):
6         super().__init__()
7         self.input_proj = nn.Linear(input_dim, d_model)
8         self.pos_embedding = nn.Parameter(
9             torch.randn(1, 30, d_model))
10        encoder_layer = nn.TransformerEncoderLayer(
11            d_model=d_model, nhead=nhead,
12            dim_feedforward=256, dropout=dropout,
```

```

13         batch_first=True)
14     self.transformer = nn.TransformerEncoder(
15         encoder_layer, num_layers=num_layers)
16     self.pool = nn.AdaptiveAvgPool1d(1)
17     head = lambda n: nn.Sequential(
18         nn.Linear(d_model, 64), nn.ReLU(),
19         nn.Dropout(dropout), nn.Linear(64, n))
20     self.direction_head = head(n_directions)
21     self.stroke_head = head(n_stroke_types)
22     self.posture_head = head(n_posture)
23
24     def forward(self, x):
25         x = self.input_proj(x) + self.pos_embedding
26         x = self.transformer(x)
27         x = self.pool(x.transpose(1,2)).squeeze(-1)
28         return (self.direction_head(x),
29                 self.stroke_head(x),
30                 self.posture_head(x))
31
32 # Parameters: 564,103
33 # Output shapes (batch=4):
34 #   Dir [4,3]   Str [4,2]   Pos [4,2]

```

Listing 1: PyTorch implementation of *TennisTransformerGPU*. The model is compact (564,103 parameters) and fully reproducible on Kaggle free-tier hardware.

4.4.2 Training Configuration. All three heads are trained jointly:

$$\mathcal{L} = \mathcal{L}_{\text{type}} \mathcal{L}_{\text{dir}} \mathcal{L}_{\text{posture}}$$

The posture head uses class-weighted loss ($2.5 \times$ upweight for “bad”). Optimizer: Adam, $\text{lr} = 5 \times 10^{-4}$, weight decay 10^{-3} . Scheduler: StepLR, $\text{step}=25$, $\gamma = 0.5$. Training: 80 epochs, batch size 16, gradient clip at norm 1.0. Hardware: Kaggle free-tier (dual T4 GPUs).

5 Experiment

5.1 Evaluation Protocol

Split 1 — Random 80/20. Strokes partitioned at the stroke level (1,024 train / 257 test). Because strokes from the same player can appear in both sets, this measures upper-bound model capacity and is included for comparability with prior work.

Split 2 — Cross-Player (Pro $\rightarrow$ Amateur). Trained on all professional strokes; evaluated exclusively on the amateur player. No training data from the test player is seen at any point. This is the more rigorous and practically relevant evaluation.

5.2 Quantitative Results

Table 4 reports per-task accuracy under both evaluation protocols, alongside majority-class baselines. For direction, the majority-class baseline is 61.3% (always predicting center); for posture it is 63.4% (always predicting good). These baselines are important context: direction accuracy of 61.9% under random split only marginally exceeds the majority-class ceiling, and the 68.3% cross-player figure reflects collapse to the majority class rather than genuine directional transfer. The stroke-type result (83.7%) and its near-zero cross-player drop (0.8%) remain the paper’s strongest finding.

Table 4: Multi-task accuracy under both evaluation protocols, with majority-class baselines.

Task	Majority Baseline	Random Split	Cross-Player
Stroke Type	77.0%	83.7%	82.9%
Direction	61.3%	61.9%	68.3% [†]
Posture	63.4%	62.6%	63.4%

[†]Collapses to majority class; genuine transfer not achieved.

5.3 Confusion Matrix Analysis

Figures 3 and 4 present per-task confusion matrices for both evaluation splits.

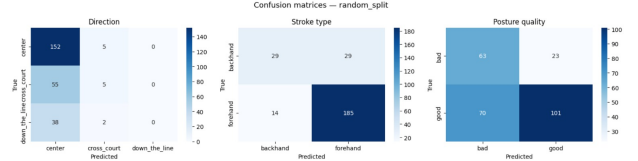


Figure 3: Confusion matrices — random 80/20 split. Left (Direction): Model defaults to center due to class imbalance. Center (Stroke Type): Strong forehand recall (185/199); lower backhand recall (29/58). Right (Posture): Moderate performance, good-posture bias.

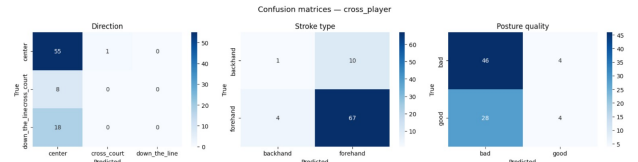


Figure 4: Confusion matrices — cross-player evaluation. Left (Direction): Collapses to majority class. Center (Stroke Type): Transfers well (82.9%). Right (Posture): Strong bad-posture recall (46/50).

5.4 Training Dynamics

Figure 5 shows validation accuracy curves across all tasks and both experimental settings over 80 training epochs.

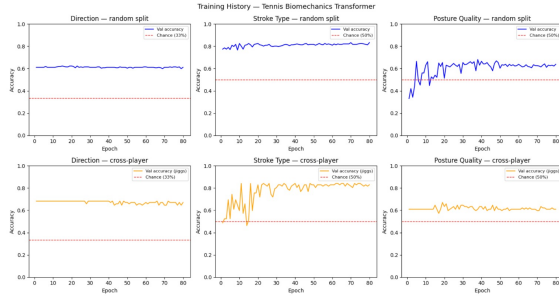


Figure 5: Validation accuracy curves — random split (top, blue) and cross-player (bottom, orange). Dashed red: chance baseline. Stroke type converges above 80% in both settings; direction plateaus near 61%; posture stabilizes around 62–63%.

5.5 Generalization Gap Analysis

Figure 6 summarizes the performance differential between random and cross-player evaluation settings across all three tasks.

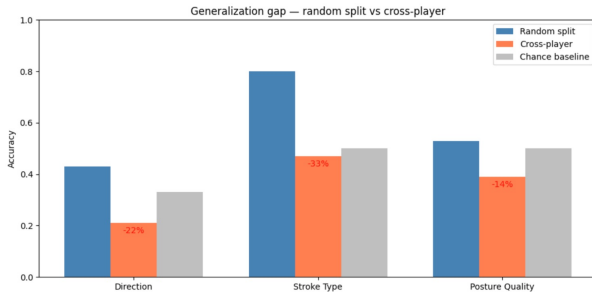


Figure 6: Generalization gap (random vs. cross-player). Stroke type shows the largest absolute drop but remains above the 50% chance baseline. All cross-player results remain above their respective chance baselines.

5.6 Comparison with Shimizu et al.

Table 5 compares our system against the baseline across pose representation, pipeline design, task coverage, and deployment. Our stroke-type accuracy (83.7%) is marginally below Shimizu et al.’s reported range ($\sim 85\text{--}90\%$), though the datasets differ substantially—their work used large broadcast footage while ours uses 11 videos collected under mixed conditions. The comparison should therefore be read as qualitative rather than a direct benchmark.

Table 5: Systematic comparison against Shimizu et al. [6].

Aspect	Shimizu et al.	Ours
<i>Pose Representation</i>		
Pose tool	OpenPose	MediaPipe Pose Landmarker (full)
Landmarks	18 (pixel only)	33 (image + world coordinates)
Coordinate space	Image (pixel, cam.-dep.)	World (metric, hip-origin, cam.-indep.)
Depth (z-axis)	No	Yes
World coords w_x, w_y, w_z	No	Yes
Visibility scores	No	Yes
Features per frame	~ 36	39
<i>Pipeline Design</i>		
Stroke segmentation	Manual / pre-annotated	Automated velocity-based
Player-adaptive thresholds	No	Yes (pros 0.07–0.08, amateur 0.04)
Model type	CNN / SVM	Transformer Encoder
Temporal modeling	Limited	Full sequence (30 frames)
Attention mechanism	No	Yes (4 heads, 4 layers)
Multi-task learning	No	Yes (2 learned tasks + rule-based feedback)
<i>Task Coverage</i>		
Stroke type classification	$\sim 85\text{--}90\%$	83.7% (random) / 82.9% (cross-player)
Direction prediction	No	Yes — 61.9% (random)
Posture quality assessment	No	Yes — 62.6% (random)
Shot selection feedback	No	Yes (rule-based, no extra labels)
Cross-player generalization	No	Yes (pro $\rightarrow$ amateur)
Amateur player support	No	Yes
<i>Dataset and Deployment</i>		
Players in dataset	Professionals only	7 pros + 1 amateur
Dataset size	Large broadcast	1,281 strokes, 11 videos
Hardware required	GPU workstation	Kaggle free tier (T4 GPU)
Real-world deployable	Yes (with hardware)	Yes (free hardware)
Reproducible (freely)	No	Yes

5.7 Qualitative Inference Results

Figures 7, 8, and 9 demonstrate system output on unseen video, showing pose overlay, per-task predictions with confidence, and rule-based shot selection feedback.

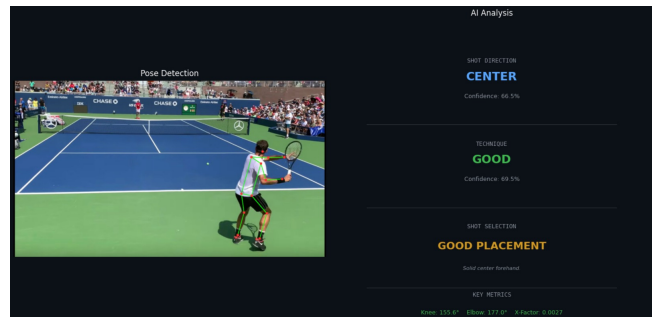


Figure 7: Inference dashboard: direction CENTER (66.5%), technique GOOD (69.5%), rule-based feedback “Good Placement.” Joint metrics: Knee 155.6°, Elbow 177.0°, X-Factor 0.0027.



Figure 8: Professional forehand (Indian Wells). Direction CENTER (56%), Stroke FOREHAND (96%), Posture GOOD (79%).

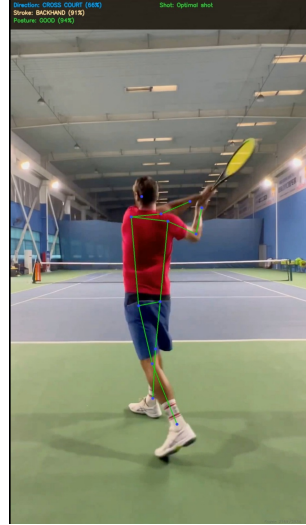


Figure 9: Amateur backhand (indoor). Direction CROSS COURT (66%), Stroke BACKHAND (91%), Posture GOOD (94%).

5.8 Discussion

Stroke type accuracy (83.7% random, 82.9% cross-player) confirms that world-coordinate features provide partially view-normalized representations sufficient for forehand/backhand discrimination. The near-zero cross-player drop suggests grip-pattern universality is well captured, though view-normalization remains incomplete for non-baseline camera angles.

Direction prediction at 61.9% barely exceeds the 61.3% majority-class baseline, indicating the model has not meaningfully learned directional intent from full-sequence features. Two factors limit this task: (1) directional cues concentrate in the 2–3 frames around ball contact, which are diluted across the full 30-frame window; (2) training data is predominantly behind-the-baseline, restricting view-invariant learning. Direction is the clearest target for future architectural work.

Posture quality at 62.6% falls just below the 63.4% majority-class baseline on random split. The more informative result is cross-player bad-posture recall of 46/50, which is the most coaching-relevant metric: flagging poor technique is more actionable than confirming good technique. Single-annotator labels remain a confound and regression-based supervision is the most direct remedy.

Cross-player generalization for stroke type (83.7% $\rightarrow$ 82.9%) is the paper’s most practically significant finding. A model trained on professional footage deploys to an amateur player with negligible accuracy loss, which is a meaningful result for low-resource coaching applications.

5.9 Limitations and Future Work

Camera viewpoint dependency. The system degrades on side/front views. Although world coordinates are nominally camera-independent, landmark visibility deteriorates in non-standard projections. Multi-angle data collection and viewpoint augmentation are the most direct remedies.

Direction model scope. Replacing full-sequence direction modeling with a contact-snapshot extractor (2–3 frames around peak wrist velocity) is the highest-priority architectural change.

Binary posture labels. Per-joint continuous scoring and multi-annotator agreement would provide richer supervision and more reliable output than single-annotator binary labels.

Future work. (1) Expand the amateur dataset to 150+ strokes; (2) implement contact-window direction modeling; (3) add multi-angle training data; (4) move to regression-based posture assessment; (5) optimize for mobile on-device inference.

6 Conclusion

We presented a multi-task tennis stroke biomechanics pipeline extending Shimizu et al. in four directions: automated velocity-based segmentation, richer 3D pose via MediaPipe world coordinates, two novel learned tasks (direction and posture) plus a rule-based feedback layer, and a cross-player generalization experiment including amateur play. *TennisTransformerGPU* (564,103 parameters) achieves 83.7% stroke-type accuracy with a 0.8% cross-player drop—the paper’s strongest result, and a practical indicator that professional training data transfers to amateur inference. Direction and posture tasks do not meaningfully exceed majority-class baselines, and both remain open problems. Contact-moment direction modeling and multi-angle robustness are the most impactful directions for improvement. The full system runs on free Kaggle hardware and is openly reproducible.

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